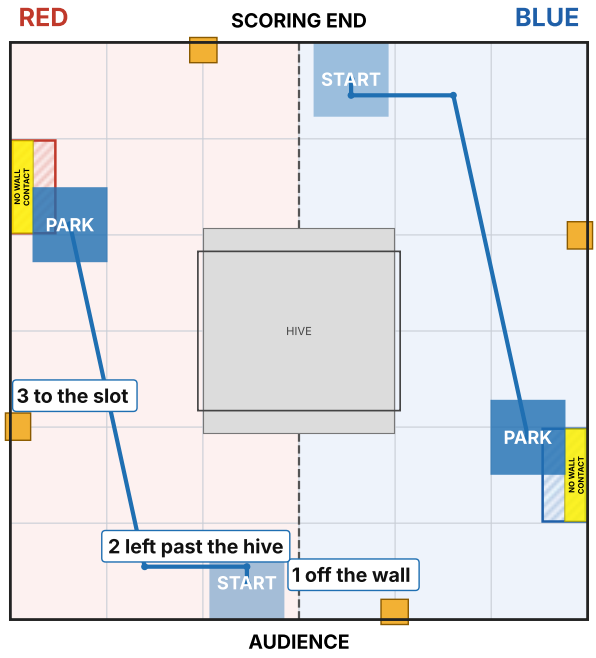
Odo ROBOT Shoot + Park

Use this when we shoot first. It is our main autonomous.

The robot shoots its 4 pollen, then drives to the parking slot on its own. It measures where it is as it drives (odometry), so it ends up in the same place every time.

What the robot does



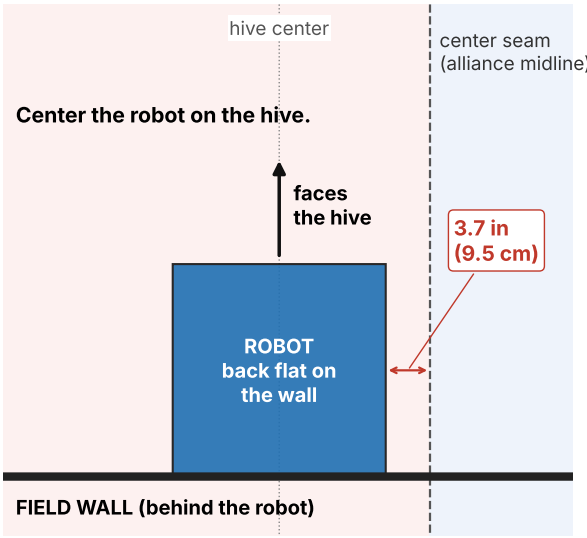
Time	What happens
0 - 0.5 s	Backs gently into the wall behind it, to line itself up straight. This is normal.
0.5 - 3.5 s	Launcher spins up.
3.5 - 13.5 s	Shoots the pollen.
about 13.5 - 19 s	Drives: (1) a little off the wall, (2) left past the hive, (3) to the **INNER** parking slot. Then it stops.

Points if it works: **LEAVE 3 + PARK 5**. Our partner parks in the **OUTER** slot.

Before the match

1. Agree with the partner. We shoot first. They shoot second (or don't shoot), stay out of our lane, and park in the OUTER slot. Show them the pit sheet.

2. Put the robot on the field.



- **Red:** at the **audience** wall. **Blue:** at the **scoring-end** wall.
- Back of the robot **flat against the wall**, facing the hive.
- **Center the robot on the hive.** Its edge is 3.7 in (9.5 cm) from the center seam.
- Exactly **4 pollen** loaded. Not touching a flower. Not in the loading zone.

3. On the Driver Hub, pick **Odo ROBOT Shoot + Park**. It's in the autonomous list, group *Pollen Odometry*.

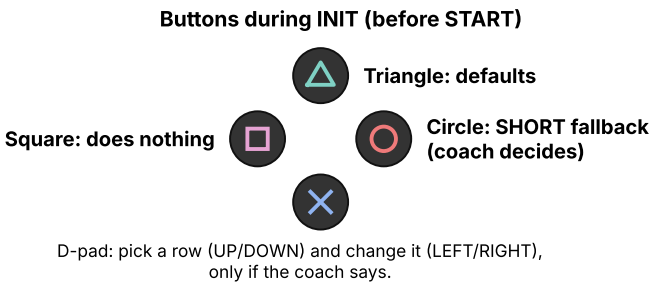
4. Press INIT. Don't touch the robot for 1 second: the odometry sensor is getting ready.

5. Press Triangle once. That puts every setting back to the defaults. Then make only the changes the coach asks for (usually none).

6. Check the screen:

Line	Should say
Build	`pollen-odo-v5`
Mode	`ODOMETRY Shoot + Park`
Odometry x fwd / y left / heading	numbers near 0 (not `NOT VALID`)
Settings notice	`Defaults: shoot, then 3 legs to the parking slot`
Status	`READY - no movement until START`

7. Press START when the match starts.



- **Triangle:** back to the defaults. Always safe.
- **Square:** does nothing. (It used to load this route; now it's the default.)
- **Circle:** the short fallback. Only if the coach says. See Leave-the-wall fallback.
- **D-pad:** changes settings. Only if the coach says.

If something is wrong

You see	What it means	Do this
`BLOCKED: ... No goBILDA Pinpoint named "odometry"`	The odometry sensor isn't found	Use **Pollen ROBOT Shoot + Park** instead (timed fallback). Tell a programmer.
`Odometry ...: NOT VALID`	The odometry sensor isn't reading	Press STOP, then INIT again with the robot still. Still bad: use the timed fallback.
`READY FOR DEPARTURE ONLY - SHOOTER FAILED`	The launcher didn't start up	The robot will still drive and park, but won't shoot. Tell a programmer; check the launcher cables.
`NOT READY - over time budget`	Someone's settings take too long	Press Triangle.
Any other `BLOCKED`	Something isn't plugged in, or the wrong robot configuration is active	Tell a programmer.

After the match, tell the programmers

- Did it shoot all 4? Did it park in the slot? Was it touching the side wall?
- Anything under **Warning** on the screen, for example `LEG 3 TO PARK gave up 4.0 in short`.
- If it stopped early: the last **Phase** shown, for example `NO ODOMETRY POSE - STOPPED` or `TIME LIMIT - STOPPED`.